a) P(t=1)

$$\alpha_1(j) = P(O_1 = o_1, X_1 = j)$$

$$P(A|B) = \frac{P(A,B)}{P(B)} \Rightarrow P(A,B) = P(A|B) \cdot P(B) \text{ def}$$

$$\alpha_1(j) = P(O_1 = o_1 | X_1 = j) \cdot P(X_1 = j)$$

$$= b_j(o_1) \cdot \pi_j$$

b) $\alpha_t(j) = P(O_1 = o_1, \dots O_t, X_t = j) =$

$$= \sum_i P(O_1 = o_1, \dots O_t, X_{t-1} = i, X_t = j)$$

$$P(A,B,C) = P(A|B,C) \cdot P(B|C) \cdot P(C)$$

$$= \sum_i P(O_t = o_t | X_t = j, X_{t-1} = i, O_{1..t-1}) \cdot \underbrace{\alpha_{t-1}(i)}$$

$$\cdot P(X_t = j | X_{t-1} = i, O_{1..t-1}) \cdot P(X_{t-1} = i, O_{1..t-1})$$

For $P(O_t = o_t | X_t = j, X_{t-1} = i, O_{1..t-1})$, the current observation $O_t = o_t$ only depends on $X_t = j$, so the rest can be removed and it becomes $b_j(o)$

For $P(X_t = j | X_{t-1} = i, O_1 \dots O_{t-1})$, the next state X_t only depends on the previous state X_{t-1}, so the old observations are dropped and it becomes a_{ij}

the sum becomes

$$\alpha_t(j) = \sum_i b_j(o_t) \cdot a_{ij} \cdot \alpha_{t-1}(i)$$

$$= b_j(o_t) \sum_i a_{ij} \cdot \alpha_{t-1}(i)$$

c) $P(o_1 = o_1, \dots o_T = o_T) = \sum_j P(o_1 = o_1, \dots o_T = o_T \mid X_T = j)$

(X_T covers everything and they don't overlap)

$$= \sum_j \alpha_T(j)$$